Class	AE460A
Semester	F2025
Group	2
Number of Students in Team	7

SOW	RCR	**Key:** P = Process and Primary or Sample Calculations R = Results	CAS/APT	SSBJ	460B	Basic Points	Content Score (0-10)	Format Score (0-10)	Points Earned	1 Max O'Neill - Project Manager	2 Ben Torres - Aero	3 Luke Horton - Performance	4 Bryce Sullins - S&C	5 Ian Plant - Structures	6 Ethan Barichievich - Subsystems	7 Willaim Peterson - Consultant
							WF 0.7	WF 0.3		Percent Participation						
			X	X												
		WEIGHT EST. OF DESIGN OPTIONS (TM-0-2)	X	X		10	10.0	10.0	100	30.0		14.0	14.0	14.0	14.0	14.0
2.2		**DESIGN AND PROGRESS REVIEWS (3)**	X	X		30	10.0	10.0	300	16.7		16.7	16.7	16.7	16.7	16.7
		OVERALL DESIGN (FALL)	X	X		40	10.0	10.0	400	16.7		16.7	16.7	16.7	16.7	16.7
		OVERALL DESIGN (SPRING)	X	X	X	40	10.0	10.0	400	16.7		16.7	16.7	16.7	16.7	16.7
			X	X												
8.7		TABLE OF CONTENTS, FIGURES, TABLES, etc	X	X		1	10.0	10.0	10	100.0						
8.7		TABLE OF SYMBOLS	X	X		3	10.0	10.0	30	100.0						
			X	X												
	1	**EXECUTIVE SUMMARY**	X	X												
	1.1.1	Description	X	X		10	10.0	10.0	100	100.0						
	1.1.2	OV-1 Diagram, ISO-view/Key Features	X	X		20	10.0	10.0	200	100.0						
	1.2	Status of Design	X	X		10	10.0	10.0	100	100.0						

2	**SYSTEM REQUIREMENTS**	X	X												
2.1	SRD Allocation	X	X		10	10.0	10.0	100	100.0						
2.2	Design Changes Required	X	X		2	10.0	10.0	20	100.0						
		X	X												
3	**AIRCRAFT DESCRIPTION**	X	X												
3.1	Preliminary Sizing	X	X												
3.1.1	Preliminary Weight Estimate (P/R)	X	X		20	10.0	10.0	200					100.0		
3.1.2	Wing Loading/Thrust sizing (P/R)	X	X		20	10.0	10.0	200					100.0		
3.1.3	Landing Gear Loads/Tire Sizing (P/R)	X	X		10	10.0	10.0	100				100.0			
3.2	Preliminary Configuration	X	X												
3.2.1	Design progression/aircraft comparision	X	X		10	10.0	10.0	100						100.0	
3.2.2	Configuration Rationale	X	X		10	10.0	10.0	100	14.3	14.3	14.3	14.3	14.3	14.3	14.3
		X	X												
3.3	Configuration and Drawings	X	X												
3.3.1	General Arrangement	X	X		50	10.0	10.0	500					50.0	50.0	
3.3.2	Drawing Table	X	X		5	10.0	10.0	50						50.0	
3.3.3	Interior (inbd profile)	X	X		50	10.0	10.0	500					50.0	50.0	
		X	X												
4	**MASS PROPERTIES**	X	X												
4.1	Detailed Weight Estimate (P/R)	X	X		20	10.0	10.0	200					100.0		
4.2	Weights and CG (Weight and Balance)	X	X		10	10.0	10.0	100					100.0		
4.3	Longitudinal CG Travel	X	X		5	10.0	10.0	50					100.0		
4.4	Moments of Inertia (460B Only)	X	X	X	30	10.0	10.0	300					100.0		

		X	X												
5	**FLIGHT LOADS (460B)**	X	X	X											
5.1	V-n Diagrams (P/R)	X	X	X	20	10.0	10.0	200	14.3	14.3	14.3	14.3	14.3	14.3	14.3
5.2	Aerodynamic Loads (P/R)	X	X	X	30	10.0	10.0	300	14.3	14.3	14.3	14.3	14.3	14.3	14.3
		X	X												
6	**STRUCTURES (460B)**	X	X	X											
6	Main Spar/Wing Structural Sizing (P/R)	X	X	X	30	10.0	10.0	300	14.3	14.3	14.3	14.3	14.3	14.3	14.3

7	**LIFT, DRAG, MOMENT**	X	X											
7.1	Airfoil Cl vs Alpha curves	X	X											
7.1.1	Wing	X	X	2	10.0	10.0	20		100.0					
7.1.2	Horizontal Stabilizer	X	X	2	10.0	10.0	20		100.0					
7.1.3	Vertical Stabilizer	X	X	2	10.0	10.0	20		100.0					
7.2	CL vs Alpha curves (Takeoff/Climb/Landing)	X	X											
7.2.2.1	CL vs Alpha (Takeoff/Climb/Landing) (P/R)	X	X	20	10.0	10.0	200		100.0					
7.2.3	CL vs Alpha curves (Takeoff/Climb/Landing)	X	X	2	10.0	10.0	20		100.0					
7.3	CL vs Alpha curves (Flight Conditions)	X	X	1	10.0	10.0	10		100.0					
		X	X											
7.4	**DRAG (Cdo)**	X	X											
7.4	Low Speed Cdo's M<Mcritical(P/R)	X	X	30	10.0	10.0	300		100.0					
7.4	High Speed Cdo's Mcritical to 1.2 (P/R)	X	X	30	10.0	10.0	300		100.0					
7.4	High Speed Cdo's - >M 1.2 (P/R)		X	30	10.0	10.0	300		100.0					
7.4.1	Cdo vs. M graph - each component	X	X	2	10.0	10.0	20		100.0					
7.4.2	Cdo vs. M graph - each configuration	X	X	2	10.0	10.0	20		100.0					
7.4.3	Table - Equivalent Area	X	X	2	10.0	10.0	20		100.0					
7.4.4	Table - Cdo	X	X	2	10.0	10.0	20		100.0					

		X	X											
7.5	**DRAG (K)**	X	X											
7.5	Low Speed k (P/R)	X	X	2	10.0	10.0	20		100.0					
7.5	High Speed k (P/R)	X	X	5	10.0	10.0	50		100.0					
7.5	k vs M graph	X	X	2	10.0	10.0	20		100.0					
		X	X											
	DRAG POLARS	X	X											
7.6.1	CD vs CL Graph (All)	X	X	2	10.0	10.0	20		100.0					
7.6.2	CD vs CL Graph (SSBJ)		X	2	10.0	10.0	20		100.0					
7.6.3	CD vs CL Graph (CAS & APT)	X		2	10.0	10.0	20		100.0					
7.6.4	CD vs CL Graph (CAS & APT)	X		2	10.0	10.0	20		100.0					
7.6.5	CD vs CL Graph (All)	X	X	2	10.0	10.0	20		100.0					
7.6.6	CD vs CL Graph (All)	X	X	2	10.0	10.0	20		100.0					

	OEI DRAG	X	X											
7.7.1	Windmilling and Boatail Drag	X	X											
7.7.1.2	Drag (P/R)	X	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
7.7.1.3	Drag vs Velocity graph	X	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
7.7.2	Trim Drag (for OEI)	X	X											
7.7.2.2	Drag (P/R)	X	X	10	10.0	10.0	100	14.3	14.3	14.3	14.3	14.3	14.3	14.3
7.7.2.3	Drag vs Velocity graph	X	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
		X	X											
7.8	**MOMENTS**	X	X											
7.8.2	CL vs CM (Takeoff/Climb/Landing)	X	X											
7.8.2.1	CL vs CM (Takeoff/Climb/Landing) (P/R)	X	X	10	10.0	10.0	100		100.0					
7.8.3	CL vs CM (Takeoff/Climb/Landing)	X	X	2	10.0	10.0	20		100.0					
7.8.4	CL vs CM (Flight Conditions)	X	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3

		X	X											
		X	X											
7.9	**DRAG SUMMARY**	X	X	2	10.0	10.0	20		100.0					
		X	X											
8	**PROPULSION**	X	X											
8.1	Engine Selection (P/R)	X	X	2	10.0	10.0	20	100.0						
8.2	Inlet Sizing	X	X	2	10.0	10.0	20						100.0	
8.3	Integration Design Drawing	X	X	10	10.0	10.0	100						100.0	
8.4	Performance Data	X	X	2	10.0	10.0	20						100.0	

9	**PERFORMANCE**	X	X											
9.1	CC Endurance Mission (SSBJ)		X											
9.1.1	Climb		X											
9.1.1	Rate of Climb- (P/R)		X	10	10.0	10.0	100							
9.1.1.1	Rate of Climb (multiple graphs)		X	5	10.0	10.0	50							
9.1.1.2	Rate of Climb max		X	2	10.0	10.0	20							
9.1.1.3	Climb Profile-Graph		X	2	10.0	10.0	20							
9.1.1.4	Climb Profile-Table		X	2	10.0	10.0	20							
9.1.2	Cruise (Range - V/Tc)		X											
9.1.2	Cruise (Range and Endurance)- (P/R)		X	10	10.0	10.0	100							
9.1.2.1	V/Tc (multiple graphs)		X	5	10.0	10.0	50							
9.1.2.2	V/Tcmax vs altitude		X	2	10.0	10.0	20							
9.1.2.3	V/Tcmax vs weight		X	2	10.0	10.0	20							
9.1.2.4	1/Tc (multiple graphs)		X	2	10.0	10.0	20							
9.1.2.5	1/Tcmax vs altitude		X	2	10.0	10.0	20							
9.1.2.6	1/Tcmax vs weight		X	2	10.0	10.0	20							
9.1.2.7	1/Tcmax vs weight @ SL		X	2	10.0	10.0	20							
9.1.3	Mission Profile Table		X	2	10.0	10.0	20							
9.1.3	Mission Profile Graph		X	2	10.0	10.0	20							
			X											
9.2	CC High Speed Mission (SSBJ)		X											
9.2.2	Cruise (Range - V/Tc)		X											
9.2.2.1	V/Tc (multiple graphs)		X	2	10.0	10.0	20							
9.2.2.2	V/Tcmax vs weight		X	2	10.0	10.0	20							
9.2.3	Mission Profile Table		X	2	10.0	10.0	20							
9.2.3	Mission Profile Graph		X	2	10.0	10.0	20							

9.3	FC (CAS and APT)	X											
9.3.1	Climb	X											
9.3.1	Rate of Climb-(P/R)	X	10	10.0	10.0	100	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.3.1.1	Rate of Climb (multiple graphs)	X	5	10.0	10.0	50	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.3.1.2	Rate of Climb max	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.3.1.3	Climb Profile-Graph	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.3.1.4	Climb Profile-Table	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.3.2	Cruise (Range - V/Tc)	X											
9.3.2	Cruise (Range and Endurance)-(P/R)	X	10	10.0	10.0	100	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.3.2.1	V/Tc (multiple graphs)	X	5	10.0	10.0	50	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.3.2.2	V/Tcmax vs altitude	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.3.2.3	V/Tcmax vs weight	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.3.2.4	1/Tc (multiple graphs)	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.3.2.5	1/Tcmax vs altitude	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.3.2.6	1/Tcmax vs weight	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.3.2.7	1/Tcmax vs weight @ SL	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.3.3	Mission Profile Table	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.3.3	Mission Profile Graph	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3

9.4	WC (CAS and APT)	X											
9.4.1	Climb	X											
9.4.1.1	Rate of Climb (multiple graphs)	X	5	10.0	10.0	50	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.4.1.2	Rate of Climb max	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.4.1.3	Climb Profile-Graph	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.4.1.4	Climb Profile-Table	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.4.2	Cruise (Range - V/Tc)	X											
9.4.2.1	V/Tc (multiple graphs)	X	5	10.0	10.0	50	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.4.2.2	V/Tcmax vs weight	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.4.2.3	V/Tc vs M @ SL	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.4.2.4	V/Tcmax vs weight @ SL	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.4.2.5	1/Tc vs M @ SL	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.4.2.6	1/Tcmax vs weight @ SL	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.4.3	Maneuver	X											
9.4.3.1	Maneuver (P/R)	X	5	10.0	10.0	50	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.4.3.2	Max Sustained n vs. M graph	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.4.3.3	Turn Radius vs. M graph	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.4.3.4	Turn Rate vs. M graph	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.4.4	Mission Profile Table	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3
9.4.4	Mission Profile Graph	X	2	10.0	10.0	20	14.3	14.3	14.3	14.3	14.3	14.3	14.3

9.5	Takeoff Analysis	X	X											
9.5	Takeoff Analysis (P/R)	X	X	20	10.0	10.0	200							100.0
9.5	Takeoff Profile Graph	X	X	2	10.0	10.0	20							100.0
		X	X											
9.6	Landing Analysis	X	X											
9.6	Landing Analysis (P/R)	X	X	10	10.0	10.0	100			100.0				
9.6.1	Approach Speed vs Weight graph	X	X	2	10.0	10.0	20			100.0				
9.6.2	Landing Distance vs Weight graph	X	X	2	10.0	10.0	20			100.0				
		X	X											
9.7	One Engine Inoperative (OEI)	X	X											
9.7.2	Planform view-FBD	X	X	2	10.0	10.0	20			100.0				
9.7.2	OEI (P/R)	X	X	20	10.0	10.0	200			100.0				
9.7.4	Graph V vs. Critical Field Length	X	X	2	10.0	10.0	20			100.0				
9.7.5	OEI ROC vs Weight-gear down	X	X	2	10.0	10.0	20			100.0				
9.7.6	OEI ROC vs Weight-gear up	X	X	2	10.0	10.0	20			100.0				
		X	X											
9.8	Performance Summary	X	X											
9.8.1	Flight Envelope	X	X	5	10.0	10.0	50			100.0				
9.8.2	Results	X	X	2	10.0	10.0	20			100.0				

	10	**STATIC STABILITY AND CONTROL**	X	X												
	10.1	Longitudinal	X	X												
	10.1.1	Horizontal Sizing - static stability	X	X		10	10.0	10.0	100				100.0			
	10.1.2	Horizontal Sizing - control	X	X		10	10.0	10.0	100				100.0			
	10.2	Directional	X	X												
		Vertical Sizing - static stability	X	X		10	10.0	10.0	100				100.0			
		Vertical Sizing - control	X	X		1	10.0	10.0	10				100.0			
	10.3	Roll	X	X												
		Aileron Sizing - control	X	X		10	10.0	10.0	100				100.0			
			X	X												
	10.4	**DYNAMIC STABILITY (AE460B)**	X	X	X											
		Longitudinal	X	X	X											
		Dimensionless Derivatives	X	X	X	30	10.0	10.0	300	14.3	14.3	14.3	14.3	14.3	14.3	14.3
		Results	X	X	X	10	10.0	10.0	100	14.3	14.3	14.3	14.3	14.3	14.3	14.3
		Lateral	X	X	X											
		Dimensionless Derivatives	X	X	X	30	10.0	10.0	300	14.3	14.3	14.3	14.3	14.3	14.3	14.3
		Results	X	X	X	10	10.0	10.0	100	14.3	14.3	14.3	14.3	14.3	14.3	14.3
			X	X												
	11	**LIFE CYCLE COSTS**	X	X												
	11	Life Cycle Costs (P/R)	X	X		10	10.0	10.0	100							100.0
	11	Tables/Graphs	X	X		3	10.0	10.0	30							100.0
			X	X												
2.3.3	H.1	**TECHINCAL MEMO - 1**	X	X		10	10.0	10.0	100	100.0						
2.3.3	H.2	**TECHINCAL MEMO - 2**	X	X		10	10.0	10.0	100			100.0				
2.3.3	H.3	**TECHINCAL MEMO - 3**	X	X		10	10.0	10.0	100		100.0					
2.3.3	H.4	**TECHINCAL MEMO - 4**	X	X		10	10.0	10.0	100				100.0			
2.3.3	H.5	**TECHINCAL MEMO - 5**	X	X		10	10.0	10.0	100				80.0	10.0	10.0	
2.3.3	H.6	**TECHINCAL MEMO - 6**	X	X		10	10.0	10.0	100						100.0	
2.3.3	H.7	**TECHINCAL MEMO - 7**	X	X		10	10.0	10.0	100			60.0			40.0	
2.3.3	H.8	**TECHINCAL MEMO - 8**	X	X		10	10.0	10.0	100					100.0		
2.3.3	H.10	**TECHINCAL MEMO - 10**	X	X		10	10.0	10.0	100							100.0
7, 8		**OVERALL CONTENT & FORMAT**	X	X		20	10.0	10.0	200	25.0	12.5	12.5	12.5	12.5	12.5	12.5

2.4	**PEER REVIEW (FALL)**	X	X	50	10.0	10.0	500	14.3	14.3	14.3	14.3	14.3	14.3	14.3
2.5	**PRELIMINARY DESIGN REVIEW (SPRING)**	X	X	100	10.0	10.0	1000	14.3	14.3	14.3	14.3	14.3	14.3	14.3
		X	X											
	TOTALS	X	X	1310			13100							
		X	X											
	Individual Memos (added to "Individual Points Earne	X	X											
	TECHINCAL MEMO - 0-1-1	X	X	5	10.0	10.0								
	TECHINCAL MEMO - 0-1-2	X	X	5	10.0	10.0								
	TECHINCAL MEMO - 0-1-3	X	X		10.0	10.0								
	TECHINCAL MEMO - 0-1-4	X	X		10.0	10.0								
	TECHINCAL MEMO - 0-1-5	X	X		10.0	10.0								
	TECHINCAL MEMO - 0-1-6	X	X		10.0	10.0								